Subset Sums in a Fourfold Sumset

Core cutoffs and the audited bound $N \le 92$

Sum-of-Subset research project

16 October 2026

Abstract

We continue the independently audited chain from $N \le 93$ to $N \le 92$. At cardinality 93, the paired bootstrap only guarantees positive-side size 13; we first eliminate this boundary, and use the previous exact prefix inverse already in the lattice gate. For the covering argument, positive low holes are separated into three bands before mass transport. Their endpoints are 2, 4 and 14, with respective covering caps 262, 254 and 250. This preserves the newly possible three-empty-unit family for the first band while avoiding the unnecessary transport to 14 there. A unit-coefficient propagation proves outer integrality in 2,987 of the 2,992 three-empty modes. A new core cutoff retains the second-largest element on each positive side and extends the two-tail bound from 25 to 36 cutoff pairs. Core mass lower bounds and a dense-span lower bound remove incompatible combinations of small span and many holes. A new persistent-hole argument bounds the defect directly, raising a critical pair of floors from 97 and 126 to 112 and 144. High-hole groups retain one common hole variable in every coupled row. Fresh integer certificates close the complete endpoint. The stronger archived bound 84 still awaits a complete independent replay; the present result advances the audited chain.

1 Result and inherited lemmas

For a finite real set A, write $\mathrm{FS}(A) = \{\sum_{x\in X} x : X \subseteq A\}$, including zero, and $4A = A + A + A + A$, allowing repetitions. Let N denote the maximum feasible size under $\mathrm{FS}(A) \subseteq 4A$.

Theorem 1.1. *Every feasible set has at most 92 elements. Hence* $71 \le N \le 92$.

The unchanged 71-element construction is $([-31, 31] \cap \mathbb{Z}) \cup \{\pm 93, \pm 155, \pm 217, \pm 713\}$. The inherited audited bound is $N \le 93$ from [1]. It therefore suffices to exclude exactly 93 elements. Such a set contains zero: otherwise adjoining zero preserves feasibility and produces 94 elements. No inheritance of feasibility by arbitrary deletion is used.

We use the corrected integralization, peeling, simultaneous inverse, effective-hole, mass, span and covering lemmas of [4], the sharp four-extreme and joint-covering results of [3], and the non-progression and prefix results of [2, 1]. For a positive r-set X, put $T_r = r(r+1)/2$ and

$$\mu(X) = |\mathrm{FS}(X)| - 1 - T_r, \qquad \delta_0(X) = |\{x \in X : 2x \notin R(X \cup \{0\})\}|, \qquad \rho(X) = \mu(X) + 2 - 2\delta_0(X),$$

where R uses distinct-element two-sums. The inherited general floors are $\rho \ge -8$, and $\rho \ge 0$ for $r \ge 14$. If the subset-sum support is not a progression, $\mu \ge r - 3$, $\rho \ge r - 14$ for $r \ge 6$, and $\rho \ge r - 5$ for $r \ge 17$. The last threshold is not extended to 16 here.

Let $F(r, u)$ denote the exact integer prefix floor proved in [1]: for $r \ge 17$, non-progression support and $\mu \ge u$ imply $\rho \ge F(r, u)$. Its complete formula and proof are inherited; `verification/nonap_inverse.py` is unchanged. The 2,723 base cases and 5,409 integer inverse checks are rerun. The new hole-dependent estimate below strengthens F without changing it.

2 A stricter order for the size-93 lattice gate

Remove four original extremes on either side to obtain D and peel twice to $D \supset D_1 \supset E = P \cup \{0\} \cup (-Q)$. Write

$$d = 85, \quad e = 77, \quad a + b = m = 76, \quad I = (a - 38)^2,$$
$$\Lambda = 1 + T_a + T_b = 1483 + I, \qquad \Omega = 484 - I.$$

The inherited identity is

$$\rho(P) + \rho(Q) + 2\vartheta + c + \mathcal{T} = \Omega, \qquad c = 2|2D| + 18d + 85 - |\operatorname{FS}(D)|,$$

where $\vartheta, c, \mathcal{T}$ are nonnegative and $\mathcal{T} = \kappa + \ell_1 + \ell_2 + g$. The deficient-sign test $1 + T_{77} > 24 \cdot 93 + 77$ holds. The paired bootstrap gives $q^2 \le 2536$ with even q, so $q \le 50$ and $a, b \ge 13$. If one side is 13, the other is 63 and $\Omega = -141 < -8$. This contradicts the identity using the general -8 floor on the small side and the zero floor on the large side. Now both sides are at least 14, so $\Omega \ge 0$, leaving $16 \le a \le 60$. Across all 45 orientations,

$$T_{\min(a,b)} - 2 - \Omega \ge 8 > 0.$$

The simultaneous inverse supplies a common core lattice. Define $z = h_P + h_Q - \kappa$, so $0 \le z \le \min(h_P, h_Q) \le \kappa$, with $h_X < |\operatorname{FS}(X)| - 1$.

Put $\rho = \rho(P) + \rho(Q)$, $\mu = \mu(P) + \mu(Q)$ and $C_0 = 2m + 4e + 14 - \Lambda = -1009 - I$. The two inner comparisons are

$$|\operatorname{FS}(D_1)| - 2|\operatorname{FS}(E)| \le \min\{C_0 + \Omega + \rho - \mu + 2z,\ C_0 + 3\Omega - \rho - \mu\}.$$

Since the maximum is defective, $\rho \le \mu$. Thus $z \le 14$ gives an upper bound $-497 - 2I < 0$. For $z \ge 15$, both supports are non-progressions. First use $\mu \ge 70$ and the applicable side floors $r - 5$ or, at size 16, $r - 14$. If that comparison is not yet negative, the parameters are $30 \le a, b \le 46$. Use the effective-hole witness at $\beta = 7$ on each side to obtain $u_r = r + C_7(r) + 1$, and substitute the stronger *combined* estimate

$$\rho + \mu \ge F(a, u_a) + F(b, u_b) + u_a + u_b.$$

All witness tests are exact. The resulting high-hole comparison is at most -17. At the balanced parameter, the earlier floor alone gives $443 - (66 + 310) = 67$, which fails; the prefix floor gives $443 - (206 + 310) = -73$. This use of F precedes the outer lattice argument and is valid for positive real sets, so it does not assume the conclusion. The outer comparison is

$$\Lambda - (4d - 10 + \Omega) = 669 + 2I > 0.$$

An off-lattice element would give disjoint support translates and violate the corresponding strict factor-two comparison. Hence D_1 and D lie on the core lattice. Only after this step do we use hole propagation and the sharp four-extreme lemma:

$$w \le 4z, \qquad \mathcal{T} \ge \kappa + 160 - 3z, \qquad \mathcal{T} \ge \max(54, z, 160 - 2z). \tag{1}$$

At the two size-16 boundary orientations $\Omega = 0$, so no state survives (1). The prefix inverse is therefore never called outside its proved range in the subsequent covering models.

3 A direct defect bound from persistent holes

Theorem 3.1. *Let X be a positive integer r-set with hull-hole count $2\beta < h_X < |\operatorname{FS}(X)| - 1$. Suppose*

$$\beta \ge 2, \qquad r \ge L \ge \max(17, 2\beta + 5).$$

Then $\rho(X) \ge B(r, \beta, L)$, where

$$B(r, \beta, L) = \min\{(\beta - 1)(r - L + 1),\ T_L - r - 1,\ T_L + L - 6\beta - r - 7\}. \tag{2}$$

Proof. Assume $\rho(X)$ is below all three terms. The hole condition makes X primitive: a gcd at least two gives at least $|\operatorname{FS}(X)| - 1$ hull holes. Its support is not a progression, since a primitive progression support containing zero would be a full interval. Write X_i for increasing prefixes and $y_i = |\operatorname{FS}(X_i)| - |\operatorname{FS}(X_{i-1})| - i$. We use $\rho_i \geq \rho_{i-1} + y_i - 2$ and the inherited fact that, for $i \geq 6$, $y_i = 0$ implies a progression support.

If the last progression prefix had size $t \geq L$, its next translate would be disjoint, so $y_{t+1} \geq T_t - t$. Its predecessor has $\rho_t \geq 0$; every later prefix is non-progression and has $y_i \geq 1$. Consequently

$$\rho_r \geq T_t - t - 2 - (r - t - 1) = T_t - r - 1 \geq T_L - r - 1,$$

a contradiction. All prefixes of size at least L are therefore non-progression. If every $y_i > \beta$ for $L \leq i \leq r$, telescoping from $\rho_{L-1} \geq 0$ gives $\rho_r \geq (\beta - 1)(r - L + 1)$, another contradiction. Choose $i \geq L$ with $1 \leq y_i \leq \beta$.

The small-increment inverse writes $X_i = tY$, where Y is a primitive deleted interval with at most β deletions. The scale t is an integer because X_i is integral and $\gcd(Y) = 1$. If $t > 1$, final primitivity forces a first later element off $t\mathbb{Z}$, at index k. Its support translate is disjoint, and the same estimate with $\rho_{k-1} \geq 0$ and later $y_j \geq 1$ gives $\rho_r \geq T_{k-1} - r - 1 \geq T_L - r - 1$. Hence $t = 1$. Since $i \geq 2\beta + 5$, the inherited counted-endpoint lemma applies:

$$\operatorname{FS}(X_i) = [0, S]_{\mathbb{Z}} \setminus (H \cup (S - H)), \quad H \subseteq [1, 2\beta], \quad |H| \leq \beta.$$

Suppose this description holds at size $v \geq i$, with sum S. Let x be the next element. If $x \geq S - 4\beta$, its translate supplies at least $S - 6\beta$ new points above S: if $x \leq S$, at least $x - 2\beta$ of the last x positions are present; if $x > S$, the whole translate is new. Thus $y_{v+1} \geq S - 6\beta - v - 1$. The sharp non-progression floor $\rho_v \geq v - 5$ and the remaining increments $y_j \geq 1$ give

$$\rho_r \geq S + v - 6\beta - r - 7 \geq T_L + L - 6\beta - r - 7,$$

contradicting the third term. Therefore $x < S - 4\beta$. Also $x > 2\beta$, since it exceeds the maximum of a prefix of size at least $2\beta + 5$. The two central support intervals overlap or abut; the lower endpoint holes stay H, and complement symmetry gives the upper holes. Induction preserves this description to X_r, which has at most 2β holes. This contradicts the hypothesis. □

Define $\mathcal{R}(r, z)$ as the maximum of zero and the accepted $B(r, \beta, L)$ for $2\beta < z$, searching $\beta \leq 20$ and all admissible $L \leq r$. Set

$$W(r, u, z) = \max\{F(r, u), \mathcal{R}(r, z)\}.$$

For example, when $z \geq 19$, the choices $(r, \beta, L) = (36, 8, 21)$ and $(40, 9, 23)$ give floors 112 and 144. At excess floors $u_p = 148$, $u_q = 184$, the old prefix floors were 97 and 126. The direct hole argument raises their sum by 33. The new helper `verification/nonap_hole_rho.py` checks all 10,869 searched parameter rows and enforces the strict hole threshold. No numerical optimization is used to justify this theorem.

4 Bandwise mass transport

The exact effective-hole witness is recalled to specify the finite input. For integers $r, \beta \geq 1$, $C \geq 0$, let $K_0 \geq 3$ be least with $(K_0 + 1)(K_0 - 2)/2 > C + 1$, and put

$$D_0 = C + K_0, \quad E_0 = D_0 + 1,$$
$$M_0 = \max\left(2\beta, 1 + \left\lfloor \frac{1 + \sqrt{9 + 8(E_0 + 6\beta)}}{2} \right\rfloor\right), \quad J = \max(K_0 + 1, 2\beta + 5, M_0).$$

Accept a witness only if

$$T_{M_0} - 6\beta > M_0 + 1 + E_0, \quad \beta(r - J + 1) > D_0, \quad T_{J-1} - 1 > D_0.$$

Then $2\beta < h_X < |\mathrm{FS}(X)| - 1$ implies $\mu(X) \geq r + C + 1$. The catalogue searches $17 \leq r \leq 59$, $1 \leq \beta \leq 20$, $0 \leq C \leq 600$. These limits need not be optimal; every used witness is checked independently of the search. Let $U(r, z)$ be the maximum of $r - 3$ and the accepted $r + C + 1$ with $2\beta < z$. For $z = 1, 2$ use $U = r - 3$. For positive holes, (1) gives

$$c \leq 484 - I - W(a, U(a, z), z) - W(b, U(b, z), z) - \max(54, z, 160 - 2z). \tag{3}$$

For zero holes use zero side penalties, obtaining $c \leq 324 - I$. All surviving integer states are enumerated within $0 \leq z \leq \Omega$. There are 443 low-hole states and 3,602 high-hole states.

Lemma 4.1 (Three bands)**.** *For positive low holes, the following bounds hold before transport.*

Band	*Hole range*	*Combined ρ floor*	*$c + I$ cap*	*Endpoint*
P	*1–2*	*66*	*262*	*2*
Q	*3–4*	*78*	*254*	*4*
R	*5–14*	*102*	*250*	*14*

Proof. The first floor is $(a - 5) + (b - 5) = 66$. For the second, the $\beta = 1$ witnesses and F give a combined floor at least 78 for every $17 \leq a \leq 59$; the $\beta = 2$ witnesses give at least 102 for the third. All 86 witness pairs and exact inverse parameters are recorded. Since $z \leq 14$, the last term in (3) is $160 - 2z$. Thus $c + I \leq 324 - \rho + 2z$, giving exactly the table. □

In particular, $262 \geq 3d = 255$ but $254, 250 < 255$. The three-empty family is possible only in band P and must be retained. A coarse use of $\rho \geq 66$ for all low holes would give cap 286. We do not claim fewer total roots from banding: the stronger models have 52,660 positive integer roots versus 47,212 for that weaker one-band relaxation. The advantage is the smaller transport error and the stronger valid budgets, not that comparison of root counts.

Lemma 4.2 (Endpoint transport)**.** *Fix a band with upper endpoint Z and impose its constant cap first. Every actual model in the band maps to its endpoint model by*

$$z \leftarrow Z, \qquad S_p \leftarrow S_p + Z - z, \qquad S_q \leftarrow S_q + Z - z,$$

leaving other coordinates fixed.

Proof. Write a model row as $v \cdot x \leq b$ and $t = Z - z \geq 0$. Its left side changes by $t(v_{S_p} + v_{S_q} + v_z)$. Every retained row except the upper-hole row has this coefficient nonpositive, as checked symbolically by the verifier. The upper-hole row holds by equality at Z; all coordinates remain nonnegative. The cap is already independent of z, and its right side is not recomputed after transport. The lower-hole row also weakens. Therefore all rows hold at the endpoint. The inherited endpoint binary alternatives are selected afresh using the enlarged mass endpoint. □

The largest transport increments are now 1, 1 and 9 on the three bands. Sending all positive low holes to 14 would permit increments up to 13, including the delicate newly restored three-empty family.

Let j count empty units and put $\delta = c - 85j$. The low-hole root ranges, including all retained modes, are

Regime	a range	Integer roots	Regime	a range	Integer roots
Z_0	20–56	148	P_0	22–54	132
Z_1	23–53	2,170	P_1	25–51	1,890
Z_2	26–50	14,450	P_2	29–47	10,982
Z_3	30–46	50,864	P_3	36–40	14,960
Q_0	23–53	124	R_0	23–53	124
Q_1	25–51	1,890	R_1	26–50	1,750
Q_2	29–47	10,982	R_2	30–46	9,826

Here Z denotes zero holes. This is a containing universe: every surviving actual scalar state lies in the listed range, although some listed parameters may admit a stronger scalar exclusion.

5 Recovering integrality and extending the core cutoff

Lemma 5.1 (Integral-center propagation)**.** *Suppose the group residual cap is below* $2d-1=169$*. Start from the two leading centers, all selected nonempty ordinary centers, and the actual member of any split pair. Take differences of known integral forms and repeatedly solve a form with one unknown coordinate of coefficient* $+1$ *or* -1*. Every recovered coordinate is integral on the core lattice.*

Proof. Both leading losses are at most $\delta<169\le|2D|$, so each leading translate contributes. Because D and $2D$ are integral, contribution to the integral target forces its center to be integral. The same argument applies to every selected nonempty onefold translate and actual split-pair member. At most three missing units leave at least one own-center slot on each side, so both leading losses are included in their actual block deficits and are bounded by δ. Differences of integral forms remain integral. In a form $\pm x+\sum c_ix_i\in\mathbb{Z}$ with integral c_i and already integral x_i, subtraction proves $x\in\mathbb{Z}$; induction proves every step. Division by a coefficient other than $+1$ or -1 is not allowed. □

The checker begins with $c_p=p_1-p_2-p_3-p_4$, $c_q=q_1-q_2-q_3-q_4$. For every selected center v it also forms $v-c_p$ or $v-c_q$. For example, with branch A and missing labels $p0,p1,q5$, the retained $p2,p3$ centers recover p_3,p_4. The center $p8=p_2-p_3-p_4$ then recovers p_2, and the leading identity recovers p_1. The other side follows from its retained own centers. All eight coordinates are integral, so every original outer gap is at least one; in branch B its strictly positive integral sum is at least one.

Complete propagation succeeds in 2,987 of the 2,992 three-empty modes. The remaining five retain weak outer gaps. They all lie in branch B:

Missing labels	Selected second-pair member	Unrecovered coordinates
$p0,p1,p8$	p	p_1,p_2
$p1,p2,q6$	p	p_2,p_3
$p1,p3,q7$	p	p_2,p_4
$p6,q1,q2$	q	q_2,q_3
$p7,q1,q3$	q	q_2,q_4

This list describes what the stated propagation rule proves, not a claim that the five exceptional modes must have nonintegral outer elements. The verifier recomputes all modes and each unit-coefficient derivation; no integrality declaration is read from a certificate.

The decisive geometric refinement is to retain C_p, the second-largest element of P, and C_q, the second-largest element of Q. On side p, with $x_0=R_p$ and $x_4=M_p$ as below, actual sets satisfy

$$C_p\ge a-1,\qquad C_p+1\le R_p,\qquad C_p+R_p+x_1+x_2+x_3+M_p-S_p\le -T_{a-2}. \tag{4}$$

The last row counts the other $a-2$ distinct positive integers at their minimum possible sum. It remains valid after mass transport, which only increases S_p. Reflect for side q. We also use the upper mass constraint

$$S_p\le x_1+x_2+x_3+M_p+R_p+(a-1)C_p-T_{a-2}. \tag{5}$$

Indeed, the other $a-1$ core elements are distinct integers at most C_p. To keep a constant matrix over an a-interval, the verifier replaces $a-1$ in its coefficient by the group's largest possible $a-1$; this weakens the upper bound since $C_p\ge 0$. For a positive low band $[z_0,Z]$, use the invariant row obtained by subtracting z on the left and z_0 on the right. It holds before transport because $z\ge z_0$, and S_p-z is unchanged by transport. At the endpoint it allows only the justified extra mass $Z-z_0$. High groups retain the original row and their exact fixed side size.

Lemma 5.2 (Independent core cutoffs)**.** *Put* $x_{-1}=C_p$*,* $y_{-1}=C_q$*. For every* $-1\le i,h\le 4$*,*

$$|2D|\le M_p+M_q+x_i+y_h+1+T_{4-i}+T_{4-h}. \tag{6}$$

Proof. The central integer interval $[-M_q - y_h, M_p + x_i]$ has $M_p + M_q + x_i + y_h + 1$ points. A sum above its upper endpoint must use two positive elements strictly above x_i: otherwise one summand is at most x_i and the other at most M_p. There are $4 - i$ such elements, so there are at most T_{4-i} unordered pairs allowing repetition. Likewise at most T_{4-h} distinct sums lie below the lower endpoint. Adding these three bounds proves the claim, independently for each cutoff. □

The old model used only $0 \le i, h \le 4$. The added positions give 36 pairs, including 11 new rows, and two new coordinates constrained by (4). For illustration, the previously surviving size-93 relaxation at $a = 37$, $b = 39$ had

$$R_p = 73,\ R_q = 48,\ M_p = 200,\ M_q = 110,\ S_p = 1165, \quad (x_1, x_2, x_3) = (74, 75, 77).$$

Its mass row forces $C_p \le 36$. The new cutoff $(i, h) = (-1, 0)$ therefore gives $|2D| \le 200 + 110 + 36 + 48 + 1 + 15 + 10 = 420$, instead of the relaxed value 447. This is an excluded relaxation point, not a construction. The retained inner-loss disjunctions now extend to the six largest magnitudes $M_p, x_3, x_2, x_1, R_p, C_p$, and similarly on side q. If fewer than i points are lost at the lower endpoint, a contributing center is at least one above the ith negative magnitude; the upper version uses the ith positive magnitude and the mass endpoint. Thus the new two positions use the same counting proof as the first four.

For a finite sanity check, take each six-element subset of $\{1, 2, 3, 40, 41, 42, 43, 100\}$. For all 784 ordered pairs, independently compute the actual twofold sumset and check all 36 bounds and the counts outside their central intervals: 28,224 checks, with 159 strict improvements over the old cutoff minimum. The arbitrary-set statement is proved above; these finite checks do not replace that proof.

6 Dense spans and the high-hole models

Lemma 6.1 (Holes require span)**.** *Let X be a positive integer r-set, $r \ge 4$, with maximum R and h holes in its subset-sum hull. If $0 \le z \le h$, then*

$$2R \ge 2r + \min(z, r - 3). \tag{7}$$

Proof. If $2R \ge 3r - 3$, the conclusion is immediate. Otherwise integrality gives $2R \le 3r - 4$. Put $q = R - r$; there are exactly q deletions from $[1, R]$. Lev's central-interval theorem [5, Theorem 1], already used in the inherited hole lemmas, applies under $R \le 3r/2 - 2$ and gives $[2q + 1, \sum X - (2q + 1)] \subseteq \mathrm{FS}(X)$. Each missing sum in $[1, 2q]$ is a deleted element, so there are at most q such sums. Complement symmetry gives $h \le 2q$; the two end regions are disjoint since $\sum X \ge T_r > 4q$. Hence $z \le 2(R - r)$, proving the other sufficient inequality and therefore (7). □

For the three low bands, their lower endpoints satisfy $z_0 \le r - 3$. Integrality therefore gives the transport-invariant rows $R \ge r + \lceil z_0/2 \rceil$, adding respectively 1, 2 and 3 to the former $R \ge r$ floor. For a high group with hole interval $[L, H]$, keep the endpoint floor $R \ge r + \lceil \min(L, r - 3)/2 \rceil$ and the chord of the concave function $f(z) = \min(z, r - 3)$. With $l = f(L)$, $u = f(H)$ and $H > L$, this is the integer-coefficient row

$$2(H - L)(R - r) \ge l(H - L) + (u - l)(z - L).$$

When needed, an exact binary alternative keeps either $z \le r - 3$ and $2R \ge 2r + z$, or $z \ge r - 3$ and $2R \ge 3r - 3$. Both children are checked. The chord uses the same z as every covering row, not a separately chosen extremum. The group audit freezes the target chord endpoints when comparing each source state to its containing model.

The checker directly enumerates 16,593 dense deleted intervals of sizes 6–14, verifies the hole bound, and checks 14,491 integer chord points. A non-dense even progression is a negative control against dropping the density condition. These checks supplement the general proof above.

For $z \ge 15$, the scalar cap is at most $174 < 3d$. The first pair's actual contributor fixes the direction in the inherited active-center test; reflection cannot be chosen to improve the test. There remain

1,330 states with $j = 0$, 1,535 with $j = 1$, and 24 with $j = 2$. Group by $(a, j, U(a, z), U(b, z), v_p, v_q)$, where the two W floors v_p, v_q are constant, and retain a common real variable $z \in [z_{\min}, z_{\max}]$. There are respectively 17, 45 and 5 groups. Write $E_* = 484 - I - v_p - v_q$. In addition to $\delta \le C_* - 85j$, retain

$$\delta + z \le E_* - 85j, \quad \delta \le E_* - 54 - 85j, \quad \delta - 2z \le E_* - 160 - 85j.$$

The same z occurs in all mass, span and cardinality rows. Missing integer values and real intermediate values may be included in the relaxation, but the cap and hole count may not be independently maximized. The group audit also verifies the common W floors using the prefix floor and the hole witness at the group's minimum z. It checks 947,062 row comparisons over all 2,889 retained states, using the target group's fixed row policy. The high-hole root count becomes $17 \cdot 4 + 45 \cdot 70 + 5 \cdot 578 = 6,108$.

We retain the twenty-center, eighteen-unit grammar from [3, 1]; the numbers of modes for $j = 0, 1, 2, 3$ are $4, 70, 578, 2992$. Every choice of missing units is enumerated. For a potentially split pair both actual-contributor choices are kept; positivity is used only as a necessary condition for contribution. Write $x_0 = R_p < x_1 < x_2 < x_3 < x_4 = M_p$, and similarly y_i, and $K = |2D|$. The mass rows on side p are

$$\begin{aligned} 2M_p + 6R_p - 2S_p + 2z &\le -2T_a - 12 - 2u_p, \\ M_p + 4R_p - S_p &\le -T_{a-1} - 6, \\ x_1 + x_2 + x_3 + M_p - S_p + z &\le -T_a - u_p, \\ R_p + x_1 + x_2 + x_3 + M_p - S_p &\le -T_{a-1}. \end{aligned}$$

Use $u_p = 0$ for Z, $u_p = a - 3$ in each positive low band, and the constant $u_p = U(a, z)$ in a high group; reflection supplies side q. The gap rows include $R_p \ge a$, $M_p - R_p \ge 4$ and $x_i - x_{i-1} \ge 1$. The numerical span and cardinality rows are

$$R_p + R_q + \delta - 4z \le 404 - I - 85j, \qquad S_p + S_q + \delta - 2K - 4z \le 1614 - 85j.$$

Low-band caps are $324 - I - 85j$ for Z and the corresponding table cap minus $I + 85j$ otherwise. High groups use the coupled rows above.

The group-wide residual cap C_g determines all conditional rows:

Rows	Required condition
Ordinary quantiles	$C_g < r_{\min} + 5$
Core quantiles	$C_g < 81$
Basic leading quantiles	$C_g < 169$
Nested leading quantiles	$C_g < 161$
Strict outer gaps	Inherited $j \le 2$ condition, or Lemma 5.1

The inherited strict-gap condition is $j \le 1$, or $j = 2$ with $C_g < 169$. Here $r_{\min}$ is the smallest side size in the entire group. Use the strongest justified individual type. For a positive center v, global quantiles are $v + l + M_p \ge 85$ and $v - u - M_q \le S_p - 84$. Core versions replace $(M_p, M_q, 85, 84)$ by $(R_p, R_q, 81, 80)$. Ordinary versions are $v + l \ge b + 5$ and $v - u \le S_p - a - 4$. Individual loss caps are 84 globally and 80 for core rows. In particular, Z_3 has residual cap 69 and uses core rows; P_3 has cap 7 and uses ordinary rows. For both three-empty families, Lemma 5.1 licenses strict outer gaps in the 2,987 recovered modes; the five unrecovered modes retain weak gaps. High one-empty residual caps can exceed 81; no universal core shortcut is taken. High two-empty residual caps are at most 4 and hence justify strict outer gaps, without automatically justifying ordinary quantiles.

As in [3], each block deficit O is charged once. For a block of k ordinary centers,

$$O \ge 77k - (R_p + R_q + v_{\max} - v_{\min} + 1) + \max_v (l_v + u_v).$$

For a leading slot and a nonempty subset S of its retained own-center set V, use

$$O \ge 77(|S| + 2) - 2 - p_{\max} - R_p - 2R_q + \max\{L_p, l_v + u_v : v \in S\} + \sum_{v \in V \setminus S} (l_v + u_v).$$

Also impose the individual-loss sum bound; in an own block the leading loss is added to the ordinary hull bound. These are alternative lower bounds on the same deficit, not independent gains to be added. Sum only disjoint blocks and ungrouped losses, to at most δ. Leading quantiles use $2d-1 = 169$ and $2e+7+2i = 161+2i$ under the cap conditions above. The coarse model substitutes $K \le M_p + M_q + R_p + R_q + 21$. The nested model keeps five telescoping envelope levels and all 36 two-tail inequalities (6), together with (4). The coarse model eliminates C_p, C_q and their rows; the extra core information is introduced only when the coarse contradiction fails. All coordinates are nonnegative reals. The four inherited binary alternatives remain available: core quantiles or loss at least 81; ordinary quantiles or an outside endpoint; an inner loss count or its cutoff; and both or a sole contributor in the split pair. Every binary node must certify both children. Multiple large-loss centers are allowed. High groups additionally use the density alternative proved above as an optional discovery refinement. The final certificate pool uses neither that split nor the upper-mass rows (5); those optional rows are not dependencies of the final contradiction.

7 Exact coverage and reproducibility

The checker rebuilds every root and row before reading certificate assignments. There are $120292 + 6108 = 126400$ integer root systems. The verified proof statistics are

Quantity	Count
Structural groups	14,700
Coarse / nested groups	14,518 / 182
Symbolic interval checks	14,761
Integer extremum evaluations	35,013
Integer roots requiring trees	532
Binary internal nodes / leaves	1,706 / 2,238
Maximum depth	13
Final integer leaf systems	128,106
Distinct integer vectors	2,450

For $Ax \le b$, $x \ge 0$, a nonnegative integer multiplier w with $w^T A \ge 0$ and $w^T b < 0$ is an exact contradiction. The smallest normalized gap $-w^T b / \sum w_i$ is $1/936$. Of the distinct vectors, 1,354 use a new core cutoff, 434 use a density chord, and 113 use a sixth-position inner-loss row; these counts may overlap and are recorded in `method_usage.json`. Symbolic right sides are constructed as quadratics directly; their integer extrema are checked at endpoints and at the integers around the vertex. Each integer parameter is assigned exactly once. Tree leaves are checked separately, with both children required and repeated or unproved splits rejected. Unused multipliers are rejected as well. The 31 negative controls include omission of the newly required P_3 family, omission of high two-empty modes, invalid row thresholds, unproved or universal outer integrality, an unjustified density bound, arbitrary reflection, incompatible hole/cap maxima and an attempt to exclude the next cardinality 92.

Reproduction and scope

Only Python 3.10+ and its standard library are needed for checking:

```
python -I -S -B verification/upper92.py --output .work/upper92.json
python -I -S -B verification/upper92.py --exhaustive
```

The certificate data are `data/upper92_certificates.json`. Exhaustive mode checks every integer a, every numeric row against its symbolic expression, and every multiplier pointwise. In a high group, z remains a coordinate; each exact contradiction covers its entire real interval simultaneously. The optional generator `experiments/generate_upper92_certificates.py` uses SciPy only for discovery and accepts certificates only after exact checking. Standard and optimized Python results are byte-identical. Exhaustive and interval results agree except for the mode label; regeneration

is byte-identical, and the upper-93 baseline result is unchanged. The full result, witness data, nested-group index, generation logs, hole-defect witnesses, input hashes and run receipt are saved in `results/2026-10-16/upper92/`. `verify_upper92_windows.bat` runs all ten finite continuations. These computations verify the declared finite reductions, conditional on the general mathematical lemmas proved here and in the foundation; they are not a proof-assistant formalization. The audited conclusion is $N \leq 92$. Cardinality 92 requires a new lattice and coverage analysis.

References

[1] Sum-of-Subset research project, *An integer prefix inverse and the audited bound* $N \leq 93$, `reports/upper93_from94_20261016.tex`, 16 October 2026.

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[3] Sum-of-Subset research project, *Joint covering losses and the audited bound* $N \leq 95$, `reports/upper95_from96_20261016.tex`, 16 October 2026.

[4] Sum-of-Subset research project, *The audited continuation from 115 to 103*, `reports/upper103_from115_20261014.tex`, 14 October 2026.

[5] V. F. Lev, *On consecutive subset sums*, Discrete Mathematics 187 (1998), 151–160. doi:10.1016/S0012-365X(98)80006-X.